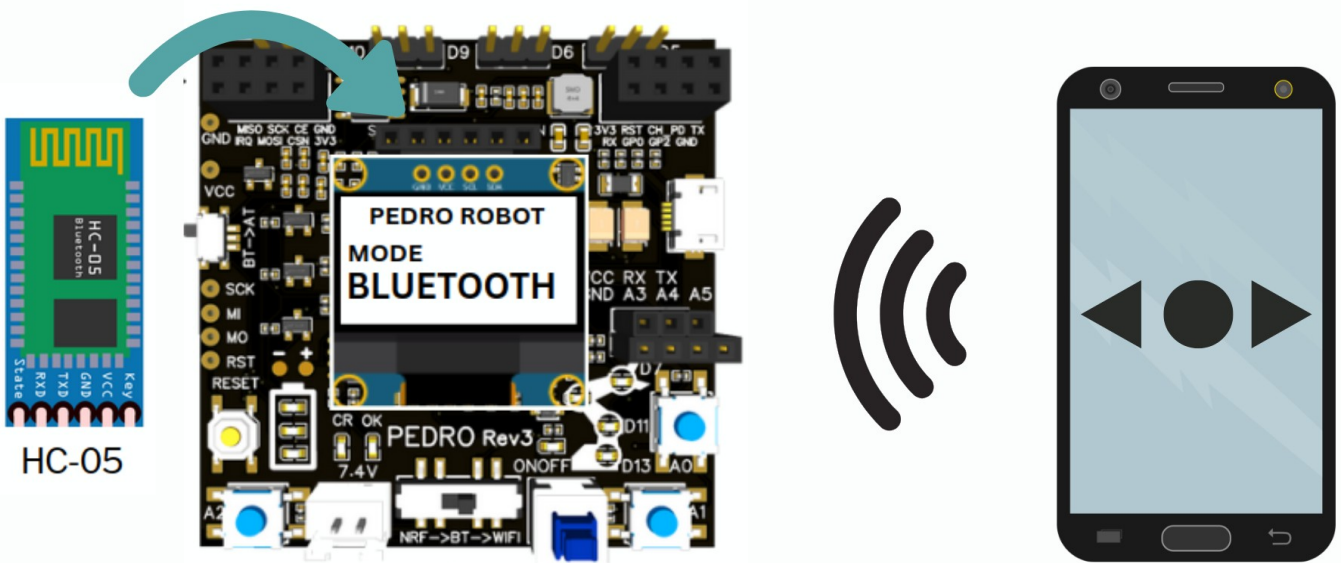


STEM Program

Lesson #8: Replay Mode



Complete this lesson to earn your Pedro Badge.

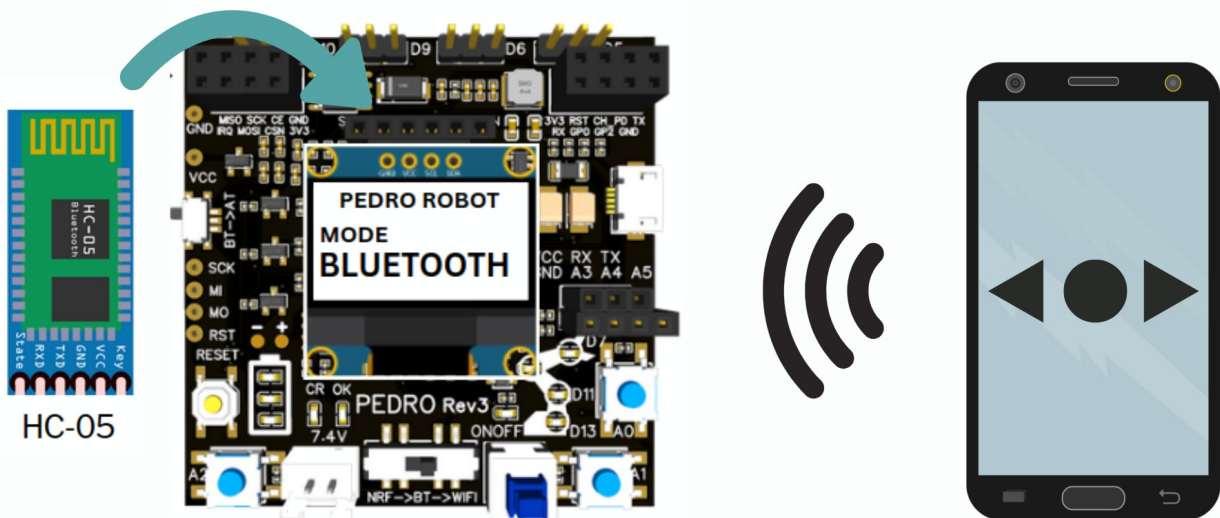


Add it to your Pedro Passport to track your STEM journey!

YOUR MISSION:

STEM Lesson n°8

Bluetooth Mode



v1.0.0

Learning Objective

In this lesson, students will discover how to control Pedro wirelessly using Bluetooth technology.

Using the integrated **HC-05 Bluetooth module**, students connect Pedro to a smartphone or tablet and send commands remotely.

By the end of this lesson, students will be able to:

- Understand Bluetooth communication
- Explain how smartphones communicate with robots
- Connect Pedro to a mobile device
- Send wireless commands to Pedro
- Understand serial communication principles

Instructor Notes

Recommended age: 12+

Difficulty Level: Beginner → Intermediate

Prerequisites: Previous Lessons

Duration: 90 – 120 minutes

 **Tip: Encourage students to predict what will happen before testing**

Required Materials

Per student or team:

- Pedro robot Assembled with HC-05 module
- Smartphone with Serial Bluetooth (available on iOS or Android)
- USB cable
- Computer with Arduino IDE installed

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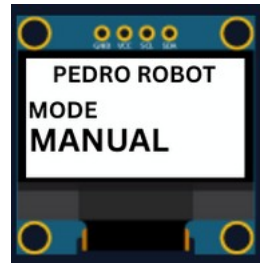
1. Install the Pedro Program

(Skip this step if already completed in Lesson nº3)

The **Pedro Program** is the core software that powers and controls the robot. It acts as the robot's "brain," managing all behaviors and interactions.

Through this single program, students can access and explore multiple control modes, allowing them to understand how a robot receives instructions, processes them, and translates them into physical actions.

- **Download and install** the latest version of the [Arduino IDE](#).
- **Install the required libraries** from the Library Manager:
 - PedroRobot: Tools → Manage Libraries → search PedroRobot → Install
 - U8glib: Tools → Manage Libraries → search U8glib → Install
 - RF24: Tools → Manage Libraries → search **RF24** → Install
- **Insert** the Oled Screen into the Pedro board and **Connect** the board to your computer via USB.
- **Select the correct port:**
 - Tools → Select the port that appear when you connect Pedro robot
- **Select the board type:**
 - Tools → Board → Arduino Micro
- **Open the example sketch:**
 - File → Examples → PedroRobot → Pedro
- **Compile and upload** the sketch to your Pedro board.

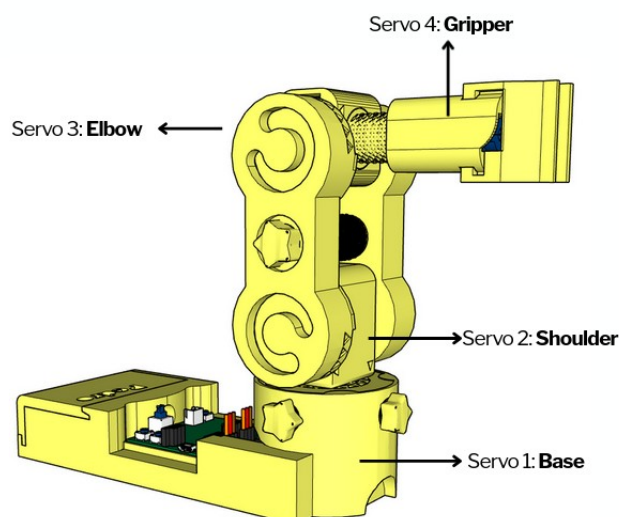


✓ If everything is correct, Pedro's OLED screen will display "MANUAL MODE".

- Press the button **A0** and observe what happen.

The LED corresponding to the selected servo light on. That button allows you to control each part of the robot. Here is the mapping between the LED ID and the servo ID

- LED D13 → servo D5 (Base)
- LED D11 → servo D6 (Shoulder)
- LED D8 → servo D9 (Elbow)
- LED D7 → servo D10 (Gripper)



2. Control Modes Overview

Using the **128x64 OLED display**, users can easily navigate through the menu to **select and configure** different modes of operation. The available control modes are:

Manual Mode – direct control using onboard buttons

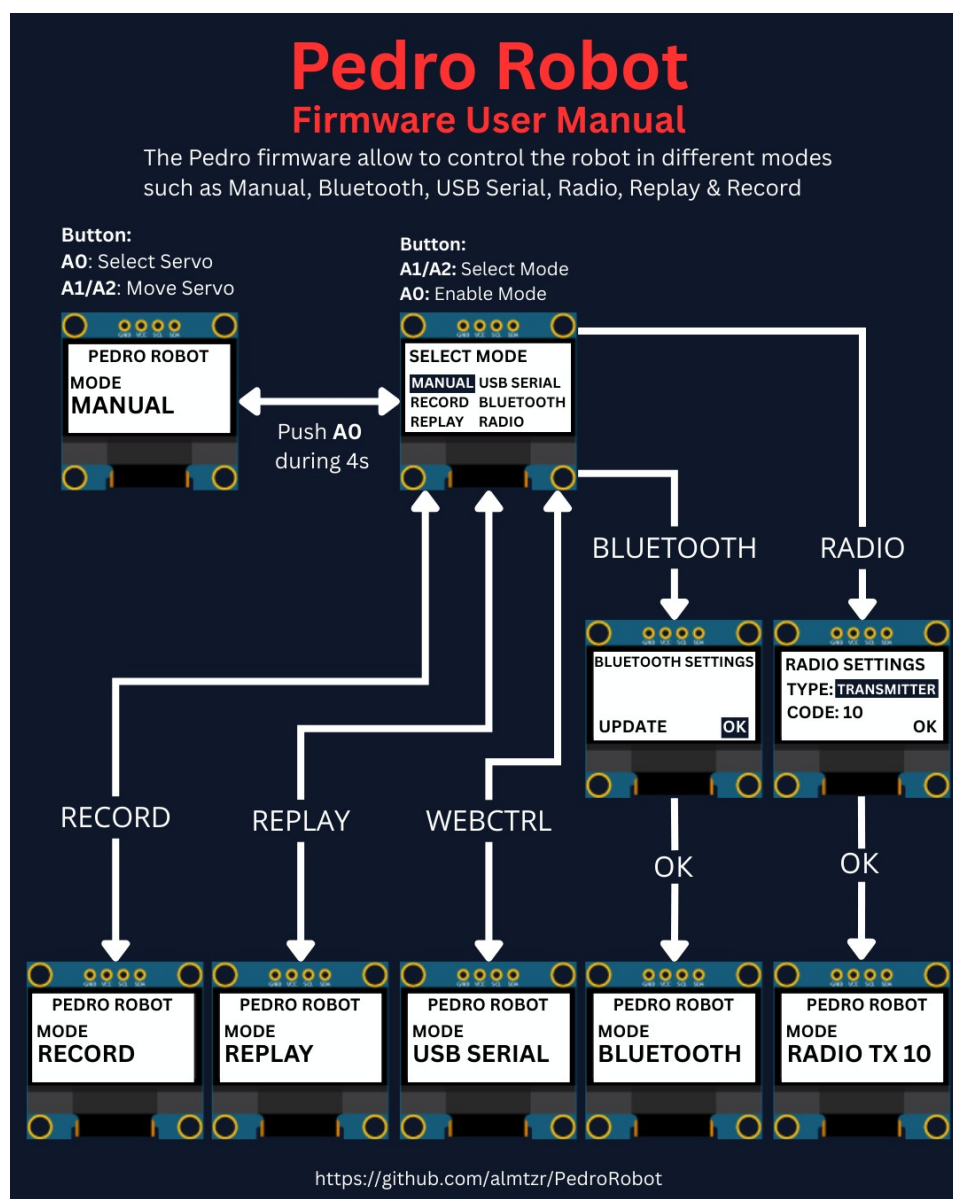
Record & Replay Mode – record and replay movements

Bluetooth Mode – control Pedro via smartphone or PC

Radio Mode – remote communication between robots

USB Serial Mode – control Pedro directly from a computer

Through this firmware, students discover how software can define robotic behavior and how multiple communication interfaces can coexist within one system. The picture below described how to navigate through the multiple control modes.



3. STEM Context

Why Bluetooth Matters in Robotics ?

Bluetooth is one of the most common wireless communication technologies. It allows devices to exchange data over short distances without cables.

Bluetooth is used in:

- wireless headphones
- smartwatches
- drones
- robots
- home automation systems

Pedro uses the **HC-05 Bluetooth module** to create a wireless communication link with a smartphone.

4. Understanding Bluetooth Communication

Bluetooth communication works like a wireless conversation. The smartphone sends digital commands. Pedro receives the commands and converts them into robot movements.

The communication chain is:

Smartphone → Bluetooth Signal → HC-05 Module → Microcontroller → Servomotors → Robot Movement

5. The HC-05 Bluetooth Module

Pedro uses an HC-05 Bluetooth module connected to the ATmega32U4 microcontroller through serial communication pins (RX/TX).

The HC-05 module:

- receives wireless data
- converts Bluetooth signals into serial data
- sends commands to Pedro's program

The HC-05 can operate in two modes:

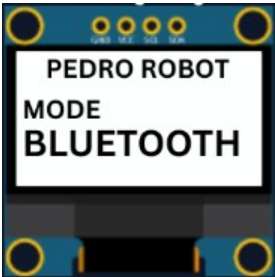
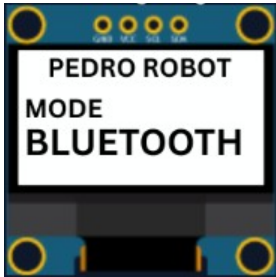
Mode	Function
Normal Mode	Standard Bluetooth communication
AT Mode	Configuration and settings

6. Entering Bluetooth Mode

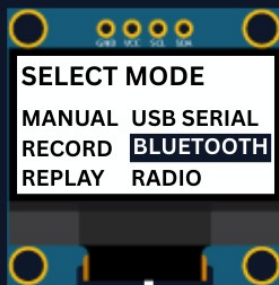
To activate Bluetooth Mode:

1. Power OFF Pedro
2. Set switch NRF → **BT** → **WiFi** to **BT**
3. Power ON Pedro (💡 Note: maintain **AT** Button to start on **AT mode** for settings options)
4. Enter the **Select Mode** menu (hold **A0** for 4 seconds)
5. Select **BLUETOOTH mode** from the menu and press **A0** to confirm
6. The OLED screen displays:



Two Setting Options	
UPDATE (AT mode required)	OK
7. When you choose UPDATE and press A0 8. The OLED screen displays:  9. That screen allows to update the name of Pedro for Bluetooth connexion 10. Set the Pedro Bluetooth name by using the button A1/A2 PEDROROBOT1 to 99 11. And click on A0 to validate 12. The OLED screen displays: 	7. When you choose OK and press A0 8. The OLED screen displays:  9. 💡 <u>Note</u>: Pedro is now ready to communicate by Bluetooth.
💡 <u>Note</u>: You can now detect the Pedro robot using your smartphone's Bluetooth search. Pedro may appear with the default Bluetooth name HC-05 or with a custom name such as PEDROROBOTn.	

Pedro Bluetooth Mode

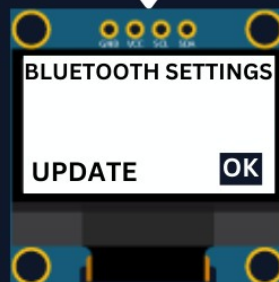


How to communicate with Pedro robot:

- Download the **Serial Bluetooth** app (iOS or Android).
- Create 7 control buttons with the following configuration:
- Name: **Servo1..4** → Value: **1..4** → Mode: **Text** → Action: **Send**.
- Name: **FwD** (Forward) → Value: **5** → Mode: **Text** → Action: **Send**.
- Name: **BwD** (Backward) → Value: **6** → Mode: **Text** → Action: **Send**.
- Name: **Stop** → Value: **7** → Mode: **Text** → Action: **Send**.

AT Mode Setup (Optional – change Pedro's Bluetooth name):

- Power **OFF** Pedro
- Set switch **NRF → BT → WiFi** to: **BT**.
- Set switch **BT → AT** to: **AT**.
- Power **ON** Pedro

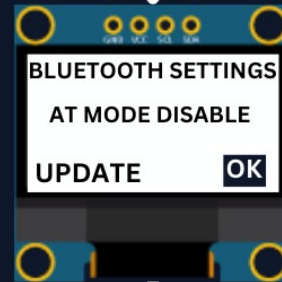


UPDATE

AT mode
Enable

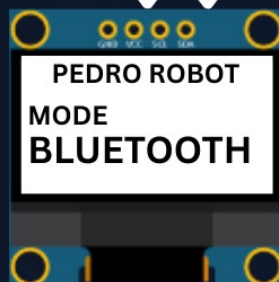
AT mode
Disable

Button A1/A2
Set Bluetooth Name
PEDROROBOT1 to 99
Button A0:
Validate



OK

OK



<https://github.com/almtzr/PedroRobot>

Bluetooth Mode Configuration Overview

7. Installing the Smartphone App

To communicate wirelessly with Pedro, students must install a **Serial Bluetooth application** on their smartphone or tablet. These applications allow the phone to send commands directly to the robot through the HC-05 Bluetooth module.

Recommended Applications

Android

- Serial Bluetooth Terminal
- Bluetooth Electronics

iOS

- Bluetooth Serial Controller

7.1 Configuring the Bluetooth Controller

After installing the application, students will create a simple control interface to operate Pedro remotely. The interface uses buttons that send text commands to the robot.

7.2 Creating the Servo Control Buttons

Create four buttons with the following configuration:

Button Name	Value Sent	Mode	Action
Servo1	1	Text	Send
Servo2	2	Text	Send
Servo3	3	Text	Send
Servo4	4	Text	Send

These buttons select which servo motor will be controlled.

7.3 Creating the Motion Control Buttons

Create three additional buttons:

Button Name	Value Sent	Mode	Action
FwD (Forward)	5	Text	Send
BwD (Backward)	6	Text	Send
Stop	7	Text	Send

These commands control the movement direction of the selected servo.

7.4 How the Communication Works

When a student presses a button on the smartphone:

1. The app sends a text command via Bluetooth
2. The HC-05 module receives the command
3. Pedro's program interprets the instruction
4. The selected servo motor moves

Example:

- Phone Button → "Servo 1" (Bluetooth command → « 1 »: the LED D13 light ON)
- Phone Button → "FwD" (Bluetooth command → « 5 »: the Servo 1 Move Forward)
- Bluetooth → HC-05
- Program reads "5"
- Servo rotates forward



Quiz — Bluetooth Mode (10 Questions)

Question 1

What module allows Pedro to communicate via Bluetooth?

- A) ESP8266
- B) NRF24L01
- C) HC-05
- D) GPS module

☒ Correct answer:

Question 2

What type of communication does Bluetooth use?

- A) Wired communication
- B) Wireless communication
- C) Optical communication
- D) Mechanical communication

☒ Correct answer:

Question 3

What device is commonly used to control Pedro in Bluetooth Mode?

- A) Printer
- B) Smartphone or tablet
- C) Keyboard only
- D) Camera

☒ Correct answer:

Question 4

What is the role of the HC-05 module?

- A) Charge the battery
- B) Receive Bluetooth commands
- C) Rotate servos directly
- D) Print 3D parts

☒ Correct answer:

Question 5

Which communication protocol is used between the HC-05 and the microcontroller?

- A) HDMI
- B) Ethernet
- C) UART Serial
- D) VGA

☒ Correct answer:

Question 6

What happens when the "FwD" button is pressed in the smartphone app?

- A) The robot turns OFF
- B) A Bluetooth command is sent to move the servo forward
- C) The OLED screen resets
- D) The battery charges

✓ Correct answer:

Question 7

Which button is used to enter the Select Mode menu?

- A) RESET
- B) A2
- C) A0
- D) POWER

✓ Correct answer:

Question 8

What is the purpose of AT Mode on the HC-05 module?

- A) Move the servos
- B) Configure Bluetooth settings
- C) Charge the robot
- D) Record movements

✓ Correct answer:

Question 9

What can students modify in AT Mode?

- A) Battery voltage
- B) OLED size
- C) Bluetooth device name
- D) Servo dimensions

✓ Correct answer:

Question 10

Why is wireless communication important in robotics?

- A) It makes robots heavier
- B) It allows robots to communicate without cables
- C) It changes the robot color
- D) It increases battery size

✓ Correct answer:

Bonus Challenge Question

How does a smartphone command become robot movement?

✓ Expected Answer:

Smartphone → Bluetooth signal → HC-05 module → Microcontroller → Servo motor → Robot movement

END

Pedro STEM Lesson nº8